**TABLE I**
## COST OF DIFFERENT ROOFTOP POWER PLANT COMPONENTS

| Component | Cost per watt (₹) |
|---|---|
| Solar modules | 32-35 |
| Inverters | 6-12 |
| Cables | 1-2 |
| Structures | 1.7-5 |
| Peripherals | 2 |
| Supervision, design and installation | 4 |

**TABLE II**
## AN ILLUSTRATION OF TAX SAVINGS FROM ACCELERATED DEPRECIATION BENEFIT

| Item | |
|---|---|
| Cost of a 100kW rooftop solar plant (A) | ₹ 6,000,000 |
| Accelerated depreciation at 40 per cent | ₹ 2,400,000 |
| Corporate tax rate | 35 per cent |
| Tax saved through depreciation (B) | ₹ 840,000 |
| **Net cost of rooftop solar plant (A-B)** | **₹ 5,160,000** |

**TABLE III**
## COST CALCULATION AFTER CAPITAL SUBSIDY

| Item | |
|---|---|
| Cost of a 100kW rooftop solar plant | ₹ 6,000,000 |
| Subsidy at 30 per cent of actual cost | ₹ 18,000,000 |
| **Net cost after subsidy benefit** | **₹ 4,200,000** |

**TABLE IV**
## COST CALCULATION AFTER INCOME TAX SUBSIDY

| Item | |
|---|---|
| Cost of a 100kW rooftop solar plant | ₹ 6,000,000 |
| Amount deducted from income | ₹ 6,000,000 |
| Tax amount saved at 30 per cent income tax | ₹ 18,000,000 |
| **Net capital cost to the installer** | **₹ 4,200,000** |

**TABLE V**
## FINAL COST OF A ROOFTOP PV SYSTEM AFTER SUBSIDIES

| Item | |
|---|---|
| Cost of a 100kW rooftop solar plant | ₹ 6,000,000 |
| Subsidy at 30 per cent | ₹ 18,000,000 |
| Net cost after subsidy | ₹ 4,200,000 |
| Accelerated depreciation at 40 per cent | ₹ 2,680,000 |
| Tax rate | 35 per cent |
| Tax saved through depreciation | ₹ 588,000 |
| **Net cost after accelerated depreciation and subsidy** | **₹ 3,612,000** |

Variable-cost heads include:
1. Human resource
2. Marketing
3. Office space

Detailed business economics, including estimated value against the above-mentioned cost heads and other probable cost heads, along with breakeven point (BEP) calculation can be worked out based on the scale of operation desired. However, an overview of solar rooftop power plant installation cost and relevant subsidies and incentives is given below.

***Solar rooftop power plant installation cost.*** Rooftop solar power plant installation costs roughly ₹ 55-60 per watt depending on the quality of components selected. Costs of different rooftop power plant components are listed in Table I. The table does not include battery backup as its cost depends on the extent of backup opted for. Not only do batteries add to the initial cost, recurring maintenance and replacement expenditure, the energy loss on charging and drawing from the battery also add to the electricity cost.

***Subsidies and incentives available for rooftop solar PV plants.*** Several incentives are available for rooftop solar PV plants. Broadly, these can be categorised as accelerated depreciation, capital subsidy and income tax credits.

Accelerated depreciation of 80 per cent was available until March 2017 under the Income Tax Act for rooftop solar PV systems, which was halved to 40 per cent from April 1, 2017 onwards. It can provide significant savings to a solar plant developer who is a taxable assesse and has sufficient profits against which the depreciation can be charged. Table II gives an illustration of tax savings from accelerated depreciation benefit.

Capital subsidy is provided in many countries for the investment made in rooftop solar power plants. Table II shows calculation for an assumed capital subsidy of 30 per cent.

Investment tax credits has a similar concept as capital subsidies. Here, instead of a direct subsidy on the capital cost, the income tax is deducted by the amount of incentive. The subsidy calculation is illustrated in Table III for an assumed income tax of 30 per cent.

Table IV shows the final cost of a rooftop PV system factoring in subsidies but without batteries. This is an estimate for commercial properties, hence includes accelerated depreciation. A similar estimation can be done for residential properties by including investment tax credits as the incentive in place of accelerated depreciation. EFY

## References


1. http://jercuts.gov.in/writereaddata/UploadFile/DPRLT_1632.pdf
2. http://npti.in/Download/MBAInternshipProject/Renewable/38_Naveen%20Kumar/PDF%20of%20Report.pdf
3. http://www.teda.in/uploads/sanction/9383195852.pdf
4. https://www.google.co.in/search?ei=qHsSWsXABcL7vATwyrTQCA&q=rooftop+solar+power+plant+project+report&oq=rooftop+solar+pv+system+project+for+home+and+office&gs_l=psy-ab.1.3.0i71k1l4.0.0.0.12931.0.0.0.0.0.0.0.0....0...1..64.psy-ab..0.0.0....0.ecFHTnYkDjY
5. Bloomberg New Energy Finance (BNEF) press releases
6. https://www.bijlibachao.com/solar/india-solar-photovoltaic-pv-panels-selection-guide-understanding-system-quality.html